

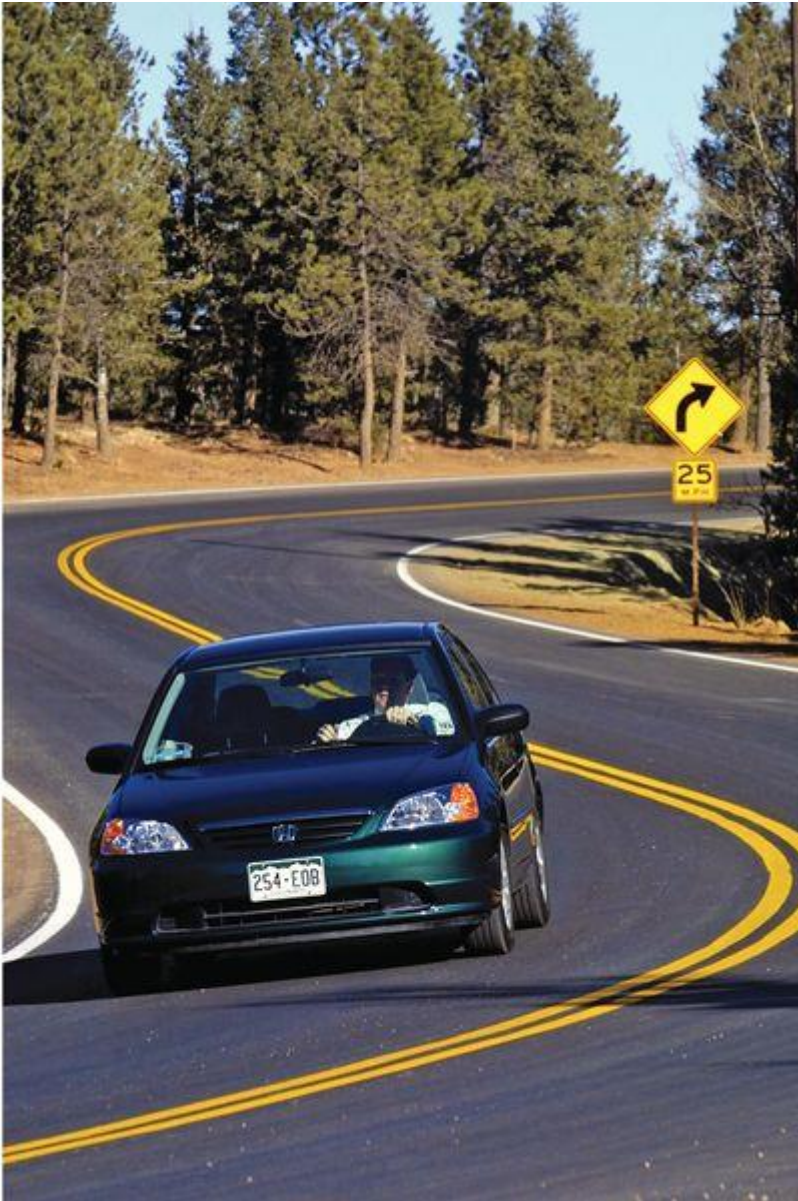
MIE 100 – LEC 0103

Lecture 4

2.4 Rectangular Coordinates (x-y)

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Lecture 4

Chapter 2: Kinematics of Particles

Howard Ho

2.1 Introduction

2.2 Rectilinear Motion

2.3 Plane Curvilinear Motion

2.4 Rectangular Coordinates (x-y)

2.5 Normal and Tangential Coordinates (n-t)

2.6. Polar Coordinates ($r-\theta$)

2.8 Relative Motion (Translating Axes)

2.9 Constrained Motion of Connected Particles



The puzzle



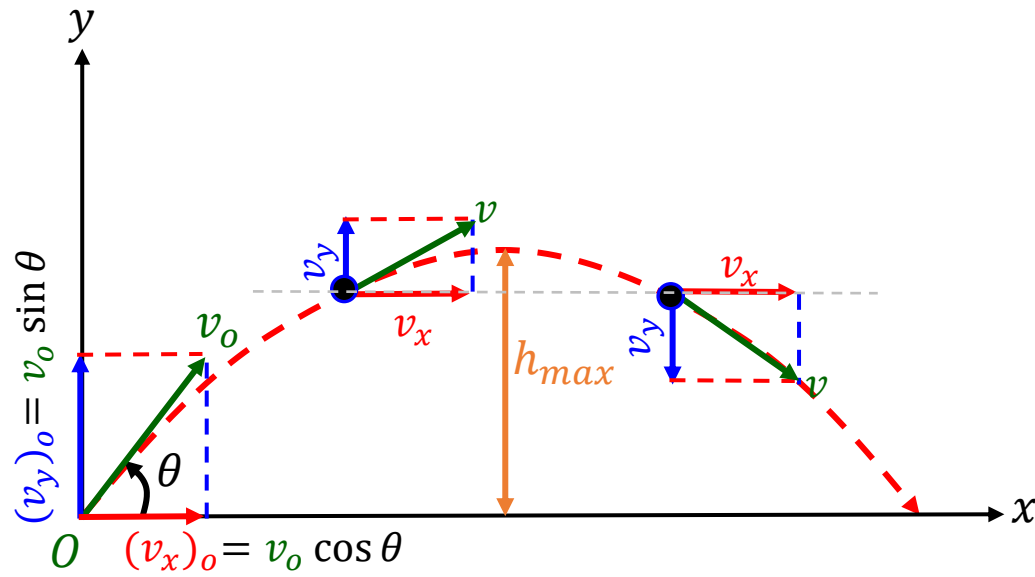
Learning Objective

In this lecture, you will learn to

- Apply the motion **equations** for analyzing **projectile motion**
- Analyze the **free-flight motion** of a **projectile particle**



Rectangular Coordinates: Projectile Motion



Assumptions

- Neglect aerodynamic drag
- Neglect curvature and rotation of earth
- Gravity is constant

- Projectile motion can be treated as **two-dimensional rectilinear motions**
- Horizontal direction

- v_x is constant, therefore acceleration is zero

$$a_x = \frac{dv_x}{dt} = 0$$

- Vertical direction

- Acceleration is constant (i.e., gravity):

$$a_y = \frac{dv_y}{dt} = a_c = -g$$

- At maximum height, h_{max} ,
 $v_y = 0$

Projectile Motion: Velocity and Position

Horizontal motion, $\vec{a}_x = 0$:

- Velocity

$$v_x = (v_x)_o$$

- Position

$$x = x_o + (v_x)_o t$$

Vertical motion, $\vec{a}_y = -g$:

- Velocity

$$v_y = (v_y)_o - gt$$

- Position

$$y = y_o + (v_y)_o t - 1/2(gt^2)$$

- Velocity-Position

$$v_y^2 = (v_y)_o^2 - 2g(y - y_o)$$

Recall: Constant Acceleration Eqns. 

$$a_c = \frac{dV}{dt} = V \frac{dV}{ds}$$

Velocity:

$$V = V_o + a_c t$$

$$V^2 = V_o^2 + 2a_c(s - s_o)$$

Position:

$$s = s_o + V_o t + \frac{a_c t^2}{2}$$

where s_o and V_o are the position and velocity at the initial time, $t = 0$

Example Problem [2.4-1]

Given: v_0 and θ , R

Find: The equation that defines y as a function of x

Assumptions:

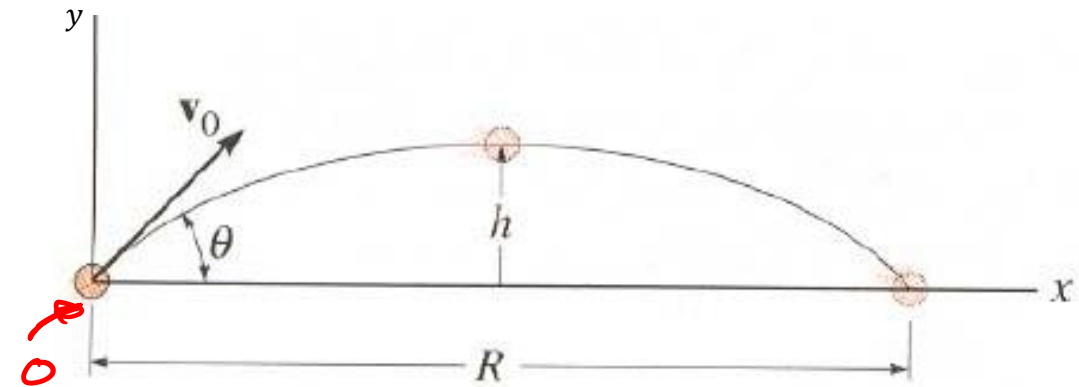
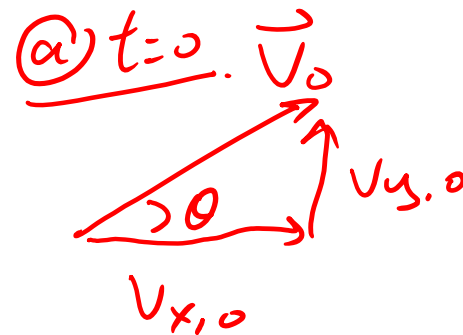
no wind drag $a_x = 0$
 $g = \text{const.}$ $a_y = -g$

At origin O , $x_0 = y_0 = 0$

Equations:

$$\cos \theta = \frac{v_{x,0}}{v_0}, \quad v_{x,0} = v_0 \cos \theta$$

$$\sin \theta = \frac{v_{y,0}}{v_0}, \quad v_{y,0} = v_0 \sin \theta$$



Special cases

($a = 0$) $s = s_0 + v_0 t$

(const a) $v = v_0 + a_0 t$

$$s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$$

$$v^2 = v_0^2 + 2a_0(s - s_0)$$

Solutions:

x-dirⁿ: (1) →

$$x = x_0 + \frac{(v_{x,0})}{\rho} t$$

$$x = (v_0 \cos \theta) t$$

$$t = \frac{x}{v_0 \cos \theta} \quad (5)$$

y-dirⁿ: (3) → $a = -g$

$$y = 0 + v_{y,0} t - \frac{1}{2} g t^2 \quad (6)$$

Sub. 5 → (6)

$$y = v_{y,0} \left(\frac{x}{v_0 \cos \theta} \right) - \frac{1}{2} g \left(\frac{x}{v_0 \cos \theta} \right)^2$$

Trig rules:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\tan^2 \theta + 1 = \sec^2 \theta = \frac{1}{\cos^2 \theta} \quad (3)$$

Special cases	
(1) →	(a = 0) $s = s_0 + v_0 t$
(2) →	(const a) $v = v_0 + a_0 t$ $s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$ $v^2 = v_0^2 + 2a_0(s - s_0)$
(4)	

Sub. $v_{y,0}$

$$y = (v_0 \sin \theta) \left(\frac{x}{v_0 \cos \theta} \right) - \frac{1}{2} g \left(\frac{x^2}{v_0^2 \cos^2 \theta} \right)$$

$$y = x \tan \theta - \frac{1}{2} \frac{g x^2}{v_0^2 (1 + \tan^2 \theta)}$$

Example Problem [2.4-2]

Given: Initial velocity v_o , launch angle θ

Find: (i) The horizontal location for placing the net

(ii) Vertical and horizontal location for ring of fire

Assumptions & Initial Conditions:

$$y_o = 10m$$

$$y_1 = 5m$$

$$v_o = 10 \frac{m}{s}$$

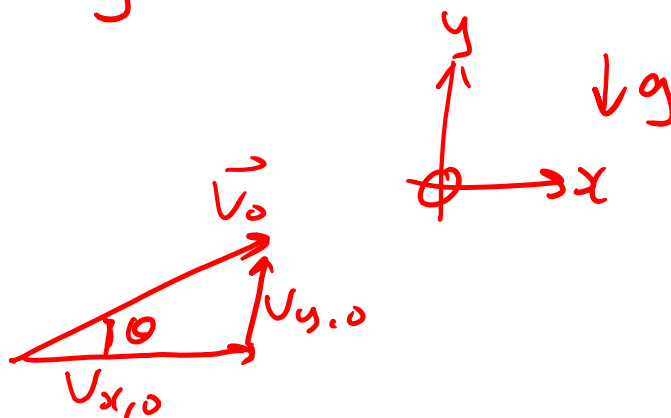
$$\theta = 30^\circ$$

Equations:

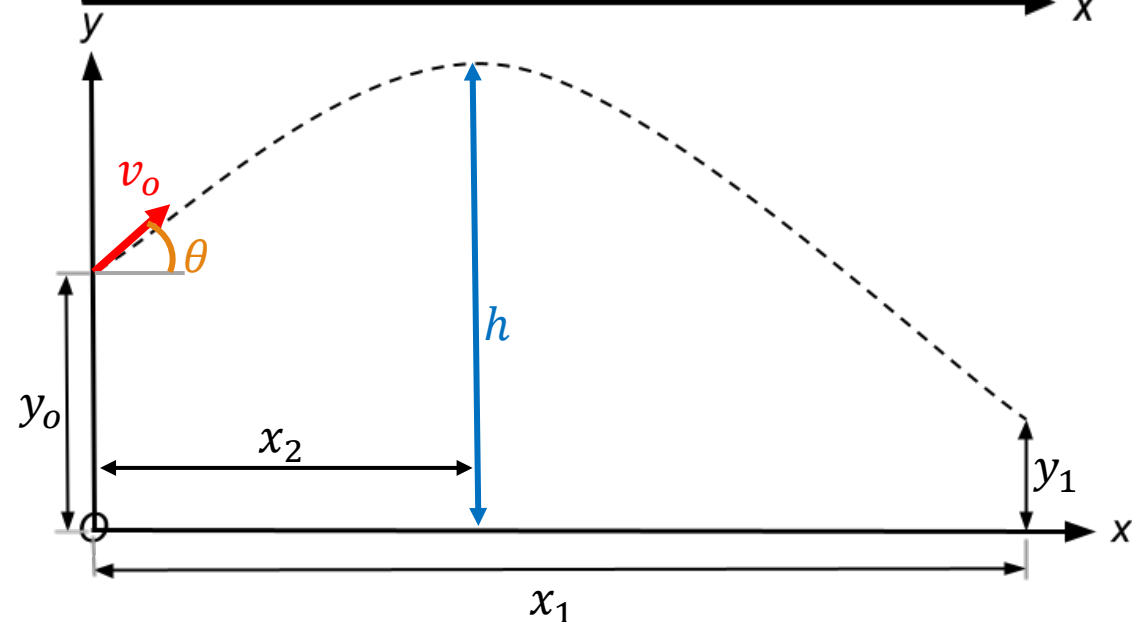
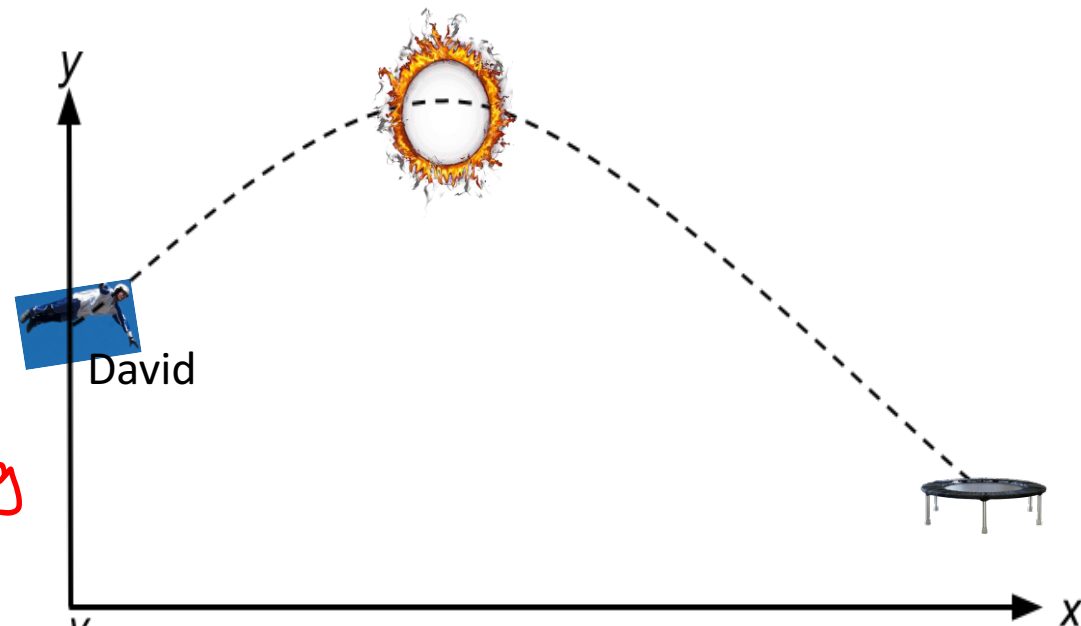
$$x\text{-dir: } v_{x,o} = v_o \cos \theta$$

$$y\text{-dir: } v_{y,o} = v_o \sin \theta$$

no wind drag $a_x = 0$
 $a = -g$



Special cases	
(a = 0)	$s = s_0 + v_0 t$
(const a)	$v = v_0 + a_0 t$
	$s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$
	$v^2 = v_0^2 + 2a_0(s - s_0)$



Solutions (i):

x-dirⁿ: (1) $\rightarrow x_1 = x_0 + (v_{x,0})t$
 $x_1 = (v_0 \cos \theta)t$ — (5)

y-dirⁿ: (3) $\rightarrow y = y_0 + (v_{y,0})t + \frac{1}{2}at^2$

$y_1 = y_0 + (v_0 \sin \theta)t - \frac{g}{2}t^2$ — (6)

Sub. I.C. into (6),

$5 = 10 + (10 \sin 30^\circ)t - \frac{9.81}{2}t^2$

$4.905t^2 - 5t - 5 = 0$

$t = 1.64s$

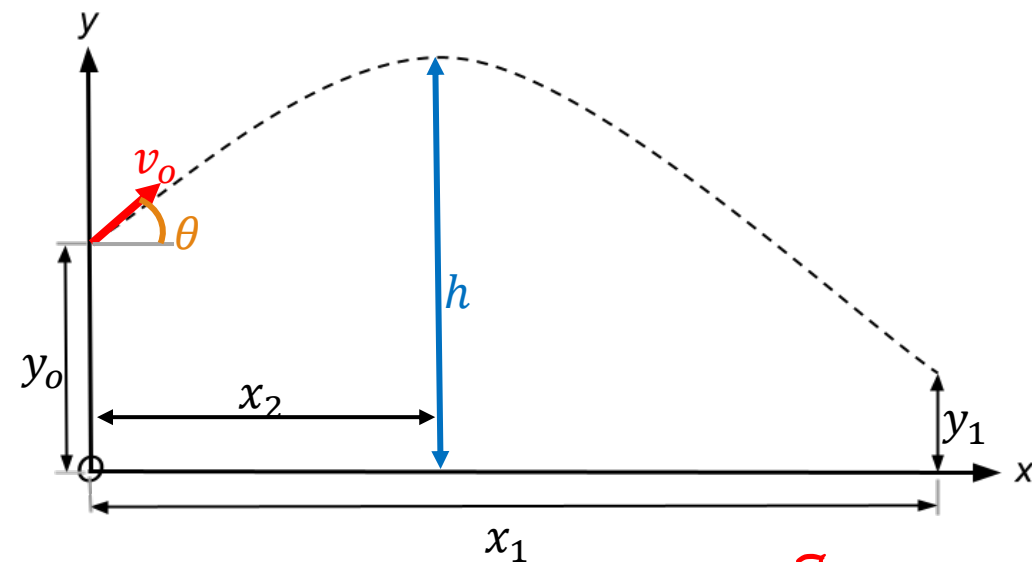
or

- 0.62s (rejected)

Sub. $t = 1.64$ into (5)

$x_1 = (10 \cos 30^\circ)(1.64)$

$\boxed{= 14.2m,,}$



I.C.

$y_0 = 10m$
 $y_1 = 5m$
 $v_0 = 10 \frac{m}{s}$
 $\theta = 30^\circ$

Special cases

- (1) — $(a = 0) \quad s = s_0 + v_0 t$
 (2) — $(\text{const } a) \quad v = v_0 + a_0 t$
 (3) — $s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$
 (4) — $v^2 = v_0^2 + 2a_0(s - s_0)$

Solutions (ii):

new I.C.

$$y = h, v_y = 0$$

$$y\text{-dir}^n: (2) \rightarrow v = v_0 + a_0 t_2$$

$$0 = (10 \sin 30^\circ) - 9.81 t_2$$

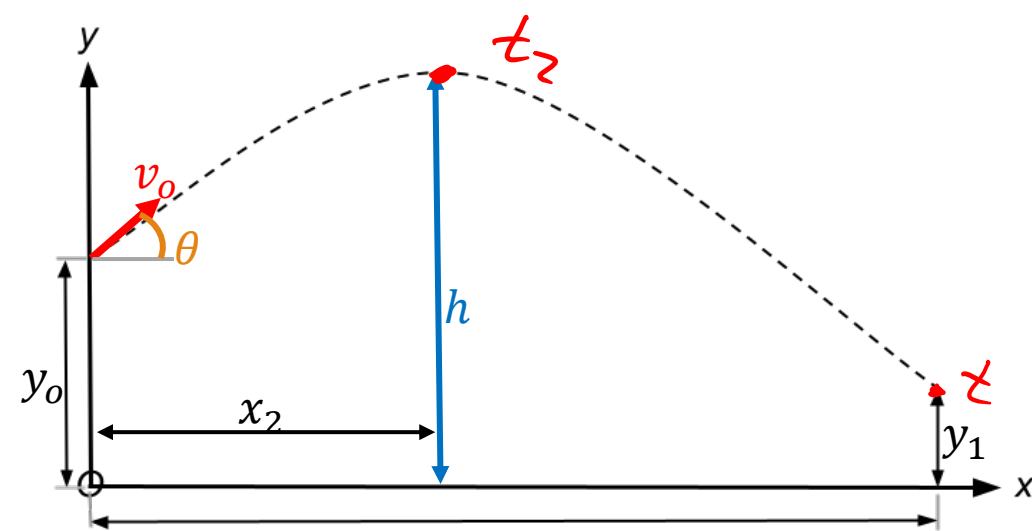
$$t_2 = 0.5097 \text{ s}$$

Sub t_2 into (3).

$$h = y_0 + (v_0 \sin \theta) t_2 - \frac{g}{2} t_2^2$$

$$h = 10 + (10 \sin 30^\circ) (0.5097) - \frac{9.81}{2} (0.5097^2)$$

$$\boxed{h = 11.27 \text{ m}}$$



$$x\text{-dir}^n: \text{Sub. } t_2 - (1)$$

$$x_2 = (v_0 \cos \theta) t_2$$

$$x_2 = (10 \cos 30^\circ) (0.5097)$$

$$\boxed{x_2 = 4.41 \text{ m.}}$$

$$y_0 = 10 \text{ m}$$

$$y_1 = 5 \text{ m}$$

$$v_0 = 10 \frac{\text{m}}{\text{s}}$$

$$\theta = 30^\circ$$

Special cases	
(a = 0)	$s = s_0 + v_0 t$
(const a)	$v = v_0 + a_0 t$
	$s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$
	$v^2 = v_0^2 + 2 a_0 (s - s_0)$

(1) —

(2) —

(3) —

(4) —

Special Cases: Horizontal Acceleration

Special cases	
(a = 0)	$s = s_0 + v_0 t$
(const a)	$v = v_0 + a_0 t$ $s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$ $v^2 = v_0^2 + 2a_0(s - s_0)$

Horizontal motion becomes, $\vec{a}_x = -a_x$:

- Velocity

$$v_x = (v_x)_0 - a_x t$$

- Position

$$x = x_0 + (v_x)_0 t - \frac{1}{2} a_x t^2$$

- Velocity-Position

$$v_x^2 = (v_x)_0^2 - 2a_x(x - x_0)$$

Solutions:

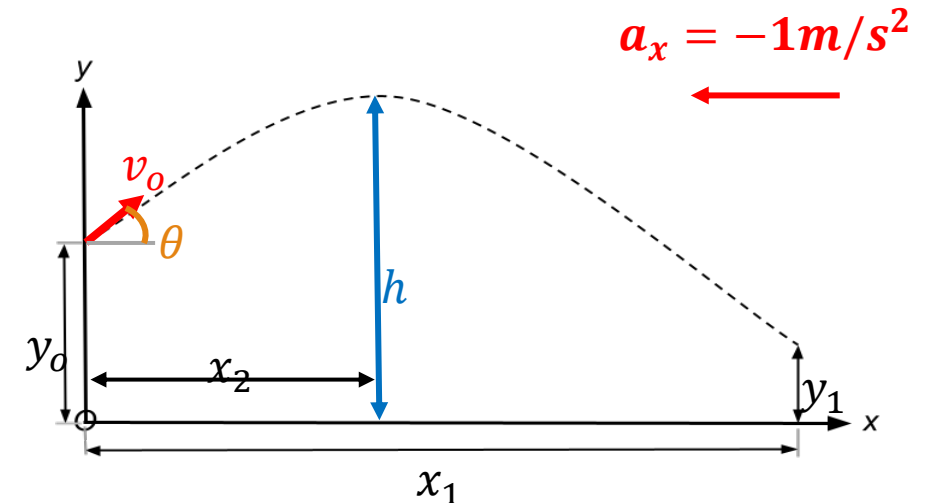
old x_1 (w/o a_x) = 14.2 m

$t_1 = 1.64$ s $a_x = -1$

$x = x_0 + v_{x,0} t + \frac{1}{2} a_x t^2$

$x = 0 + (10 \cos 30^\circ) (1.64) + (\frac{1}{2}) (-1) (1.64^2)$

$x = 12.86 \text{ m} < 14.2 \text{ m old } x_1$



In-class Exercise

The Kahoot! logo is displayed in a large, bold, white sans-serif font. The text "Kahoot!" is centered within a large rectangular area that has a dark purple background on the left and a lighter purple background on the right, separated by a diagonal line. The exclamation mark is prominent and stylized.

In-class Exercise [2.4]

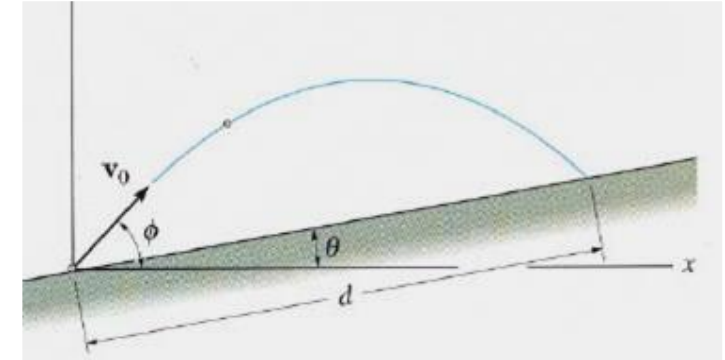
1. A projectile is given an initial velocity v_0 at an angle ϕ above the horizontal. The velocity of the projectile when it hits the slope is _____ the initial velocity v_0 .

A) greater than

C) less than

B) equal to

D) None of the above.



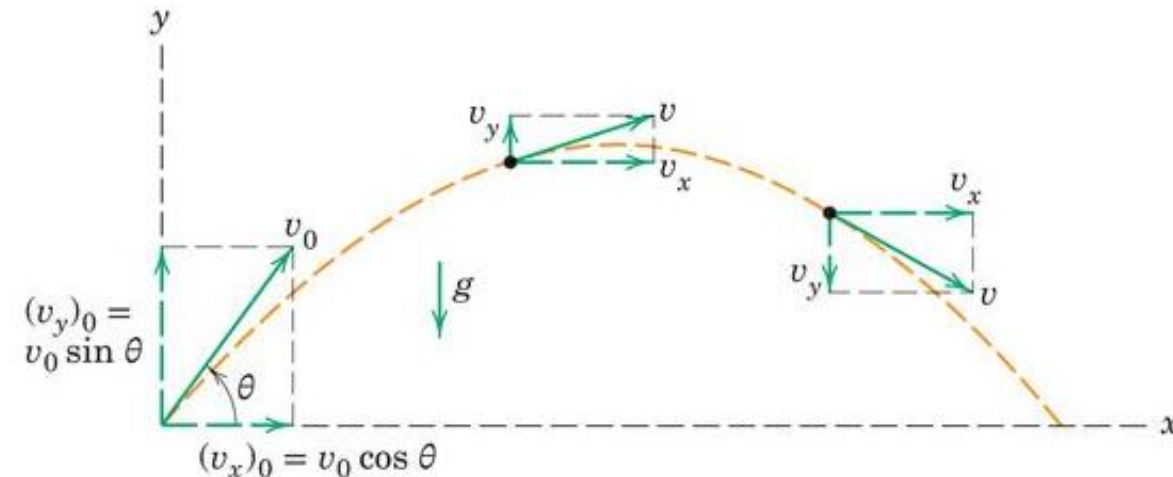
2. A particle has an initial velocity v_0 at angle θ with respect to the horizontal. The maximum height it can reach is when

A) $\theta = 30^\circ$

B) $\theta = 90^\circ$

C) $\theta = 60^\circ$

D) $\theta = 45^\circ$



In-class Exercise [2.4]

3. The time of flight of a projectile, fired over level ground with initial velocity V_o at angle θ , is equal to

A) $(v_o \sin \theta)/g$

B) $(2v_o \sin \theta)/g$ ✓

C) $(v_o \cos \theta)/g$

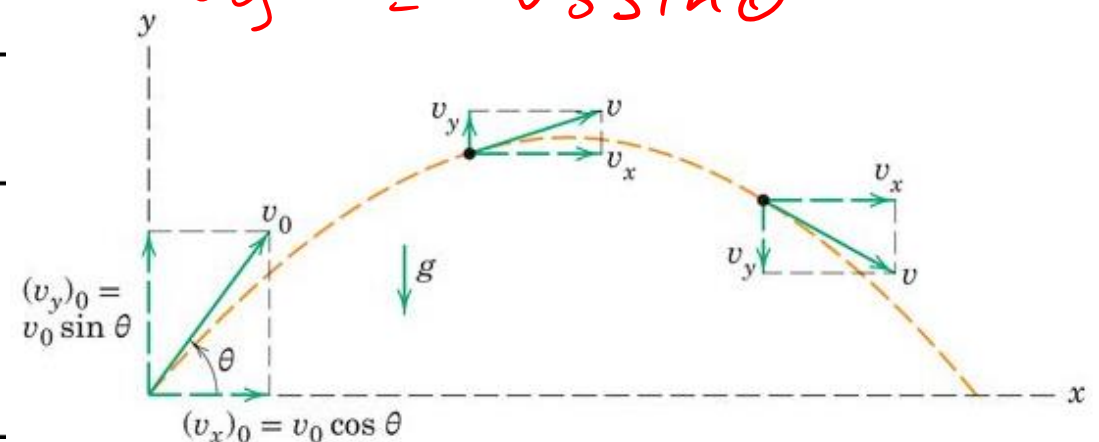
D) $(2v_o \cos \theta)/g$

✓
 $v_{y,0}$ ✓
 v_y ✓

✓
 $v_{y,0} = v_o \sin \theta$
 $v_y = -v_o \sin \theta$

✓
 $v = v_o + a_o t$
 $v_y = v_{y,0} - g t$
 $g t = 2 v_{y,0}$
 $t = \frac{2 v_o \sin \theta}{g}$

Special cases
$(a = 0) \quad s = s_0 + v_0 t$
$(\text{const } a) \quad v = v_0 + a_0 t$ $s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$ $v^2 = v_0^2 + 2 a_0 (s - s_0)$



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▼ Sample Problems

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 [Chapter 3C- Kinetics of Particles Momentum.pdf](#)

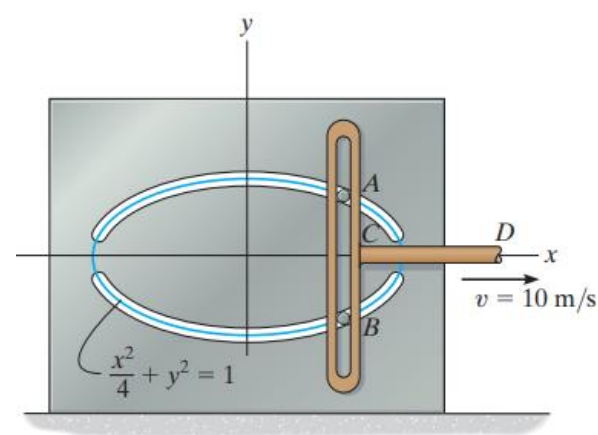
 [Chapter 5- Kinematics of Rigid Bodies Sample Problems.pdf](#)

 [Chapter 6A- Kinetics of Rigid Bodies.pdf](#)

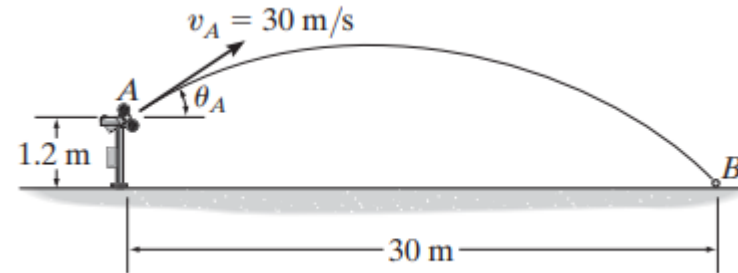
 [Chapter 6B- Work Energy Rigid Bodies.pdf](#)

12-87.

Pegs A and B are restricted to move in the elliptical slots due to the motion of the slotted link. If the link moves with a constant speed of 10 m/s, determine the magnitude of the velocity and acceleration of peg A when $x = 1$ m.

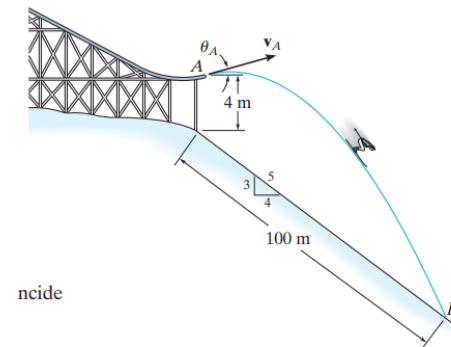


12-91. The pitching machine is adjusted so that the baseball is launched with a speed of $v_A = 30$ m/s. If the ball strikes the ground at B , determine the two possible angles θ_A at which it was launched.



12-101.

It is observed that the skier leaves the ramp A at an angle $\theta_A = 25^\circ$ with the horizontal. If he strikes the ground at B , determine his initial speed v_A and the speed at which he strikes the ground.



Learning Objective

In this lecture, you will learn to

- Apply the motion **equations** for analyzing **projectile motion**
- Analyze the **free-flight motion** of a **projectile particle**



Next lecture: 2.5 Normal and Tangential Coordinates (n-t)

